



# Silicon Info

A CPU, GPU and Neural Engine widget for Apple Silicon Macs

Documentation, version 1.1

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# 1. Overview

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Silicon Info is a small always-visible panel for the desktop. It breaks Mac activity into the pieces that make up an Apple Silicon chip instead of showing one combined number. The panel has five cards:

- **CPU** performance cores and efficiency cores, shown separately.
- **GPU** device utilization with render and tiler load.
- **Neural Engine** power draw and activity history.
- **Memory** unified memory (DRAM) usage, pressure and swap.
- **Power** how total power divides between the three blocks.

Each card is a summary. Click a card to expand it into a detailed view, and use the Back control to return. The same data is also available as a real macOS desktop widget in three sizes.

# 2. Requirements

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| Item             | Requirement                                                                      |
|------------------|----------------------------------------------------------------------------------|
| Hardware         | Mac with Apple Silicon (developed on an M4 Max)                                  |
| Operating system | macOS 14 or later                                                                |
| Build tools      | Xcode and XcodeGen ( <code>brew install xcodegen</code> )                        |
| Code signing     | An Apple Developer team. Widgets and App Groups do not work with ad hoc signing. |
| Privileges       | An administrator password, entered once per launch, for power and frequency data |

# 3. Installation and use

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## Build

Set your own team in `project.yml` ( `DEVELOPMENT_TEAM` ), and change the bundle identifiers and the App Group ( `875N49PYZ9.com.raymondmunro.siliconinfo` , in `project.yml` and `Sources/Shared/WidgetData.swift` ) to match your team. Then:

```
./build.sh
```

This generates the Xcode project from `project.yml` , then builds and signs `SiliconInfo.app` together with its widget extension.

## Run

```
open build/Build/Products/Release/SiliconInfo.app
```

The widget appears at the top right of the main display. It has no Dock icon or menu bar item. Drag it anywhere by its background. Its position is remembered, and the window is kept fully on screen when a card expands.

## Hide process names

The CPU, GPU and Memory views list process names. To hide those lists, for example before sharing your screen, start the widget with the privacy option:

```
build/Build/Products/Release/SiliconInfo.app/Contents/MacOS/SiliconInfo --hide-processes
```

## Quit

```
pkill SiliconInfo
```

# 4. The views

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## Overview

| Card          | Shows                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPU           | One bar per core, grouped into performance cores (blue) and efficiency cores (green), each group's average load, package power and a load history line. |
| GPU           | A ring for device utilization, render and tiler bars, power and a history line.                                                                         |
| Neural Engine | A ring for power draw and a history line.                                                                                                               |
| Memory        | Used and total memory, a bar split into app, wired and compressed memory, and the current pressure level.                                               |
| Power         | A single bar divided between CPU, GPU and Neural Engine, with the total in watts.                                                                       |

## CPU detail

| Section       | Contents                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Clusters      | For the efficiency cluster and each performance cluster: current frequency, active share, and a histogram of time spent at each clock step. |
| Cores         | Every core with a stacked bar of user time (blue or green) and system time (red), its current clock speed and total load.                   |
| Top processes | The busiest processes, as a percentage of one core.                                                                                         |
| Power         | Two minutes of CPU power with average and peak.                                                                                             |

## GPU detail

| Section             | Contents                                                                          |
|---------------------|-----------------------------------------------------------------------------------|
| Overview            | Utilization ring, current frequency, maximum clock, active and idle share, power. |
| Frequency residency | Time spent at each GPU clock step.                                                |
| Engines             | Device, render and tiler utilization.                                             |
| Memory              | GPU memory in use and allocated. This memory is shared with the system.           |
| Top GPU processes   | Processes using GPU time, as a share of the GPU.                                  |

## Neural Engine detail

Shows current power, average and peak over two minutes, energy used since launch, an active or idle state and a power history chart. Load is power as a share of an assumed 8 W ceiling, so treat the ring as relative.

## Memory detail

Apple Silicon uses one pool of unified memory shared by the CPU, GPU and Neural Engine, so this is the DRAM view. It needs no administrator rights.

| Section           | Contents                                                                                                                                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unified memory    | Used, available and total memory, with a bar and a legend for app memory, wired, compressed, cached files and free. Used memory follows the Activity Monitor definition: app plus wired plus compressed. |
| Pressure and swap | The system pressure level (normal, warning or critical), swap in use and swap size, and the rate of data paged in and out.                                                                               |
| Top processes     | The processes holding the most resident memory, refreshed every five seconds.                                                                                                                            |
| History           | Two minutes of memory use.                                                                                                                                                                               |

## Power detail

Shows the live split between CPU, GPU and Neural Engine, a stacked two-minute history where the top edge is the total, and average power, peak power and energy in milliwatt-hours for each block since launch.

## 5. Menu bar and desktop widget

### Menu bar

The app adds a menu bar item that shows the live total CPU load next to its icon. Its menu shows live CPU, GPU, Neural Engine and memory readings, a Hide Panel or Show Panel toggle for the floating panel, and Quit Silicon Info. If you use a menu bar manager, the icon may be inside its hidden section.

### Desktop widget

Silicon Info includes a WidgetKit widget. Run the app once, then right-click the desktop, choose Edit Widgets and search for Silicon Info.

| Size   | Shows                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------|
| Small  | One component of your choice: CPU, GPU, Neural Engine, Memory or Power. Edit the widget to pick which. |
| Medium | CPU cores, GPU and Neural Engine side by side, with memory and a power split.                          |
| Large  | Everything: CPU cores with history, GPU, Neural Engine, memory and power.                              |

### How it works

A widget runs in a sandbox and cannot read hardware counters or run `powermetrics`. The Silicon Info app therefore keeps sampling and writes a small snapshot file into an App Group container that both the app and the widget are entitled to use. The widget reads that file whenever macOS redraws it.

macOS limits how often widgets redraw, so expect an update every several seconds, not every second like the floating panel. Keep the app running for fresh data. If it is not running, the widget shows a small note asking you to open it.

## 6. Data sources

| Data                                           | Source                                                      | Needs admin |
|------------------------------------------------|-------------------------------------------------------------|-------------|
| Per-core load, user and system split           | Mach <code>host_processor_info</code>                       | No          |
| GPU utilization, render, tiler, memory         | IOKit accelerator statistics                                | No          |
| Memory breakdown, pressure, swap, paging rates | Mach <code>host_statistics64</code> and <code>sysctl</code> | No          |
| Top processes by memory                        | <code>ps</code>                                             | No          |
| CPU, GPU and Neural Engine power               | <code>powermetrics</code>                                   | Yes         |
| Core and cluster frequency, residency          | <code>powermetrics</code>                                   | Yes         |
| GPU frequency and residency                    | <code>powermetrics</code>                                   | Yes         |
| Per-process CPU and GPU time                   | <code>powermetrics</code> tasks sampler                     | Yes         |

Core order follows Apple Silicon convention: efficiency cores are numbered first, then performance cores.

## 7. Privileges and safety

Reading power and frequency needs root access. On launch the widget shows the standard macOS administrator dialog. After approval, a small shell loop runs as root. It does only one thing: it runs `powermetrics` once per second and writes the text output to `/private/tmp/siliconwidget-powermetrics.txt`, which the widget reads.

- The loop exits on its own about a second after the last Silicon Info process quits.
- Relaunching quickly reuses the running loop and does not ask again.
- Nothing is sent over the network. The output file contains only power, frequency and process names with CPU and GPU time.
- If you cancel the dialog, the widget keeps running and shows `n/a` for the values that need it.

The output file is readable by other local users, so process names on the machine are visible to them while the widget runs.

## 8. Limitations

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macOS does not expose the following on this hardware, so they are not shown:

- **Neural Engine utilization, frequency or per-process use.** Only power draw is available.
- **Per-cluster CPU power.** Only the CPU total is reported.
- **DRAM bandwidth and DRAM power.** macOS refuses the counter subscription even with administrator rights, so only memory usage is shown.
- **Media engine power.** It reads zero on this chip.

Per-process GPU time is often zero because few processes keep the GPU busy at any moment.

## 9. Troubleshooting

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| Symptom                                   | Cause and fix                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Power shows <code>n/a</code>              | The administrator dialog was cancelled. Quit and relaunch, then approve it.                                    |
| Power graphs are empty after launch       | History fills at one sample per second. Wait a minute.                                                         |
| Neural Engine reads 0 W                   | The engine is idle. Run something that uses it, such as Core ML work, to see it move.                          |
| Widget is off screen                      | Delete the saved position with <code>defaults delete com.raymondmunro.siliconinfo</code> and relaunch.         |
| Widget says "Open Silicon Info to update" | The app is not running, so no fresh snapshot is being shared. Start the app.                                   |
| Widget is missing from the gallery        | Run the app once. If it still does not appear, copy the app to the Applications folder and open it from there. |
| Build fails                               | Install the command line tools with <code>xcode-select --install</code> .                                      |

## 10. Project layout

| Path                                   | Purpose                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------|
| Sources/SiliconInfo/main.swift         | Creates the borderless floating window and keeps it anchored and on screen.                    |
| Sources/SiliconInfo/Sampler.swift      | Samples core load and GPU statistics each second and holds the history.                        |
| Sources/SiliconInfo/Memory.swift       | Reads unified memory usage, pressure, swap and the top processes by memory.                    |
| Sources/SiliconInfo/PowerFeed.swift    | Starts the privileged loop and parses the <code>powermetrics</code> output.                    |
| Sources/SiliconInfo/Views.swift        | The overview cards and expand and collapse behavior.                                           |
| Sources/SiliconInfo/Detail.swift       | The CPU, GPU, Neural Engine, Memory and Power detail views.                                    |
| Sources/SiliconInfo/WidgetBridge.swift | Builds the snapshot shared with the widget and asks WidgetKit to redraw.                       |
| Sources/SiliconInfoWidget/             | The WidgetKit extension: timeline provider and widget configuration.                           |
| Sources/Shared/                        | Data model, shared drawing code and the small, medium and large layouts, used by both targets. |
| project.yml                            | Xcode project definition, generated by XcodeGen.                                               |
| scripts/make_icon.swift                | Draws the app icon.                                                                            |
| build.sh                               | Generates the project and builds the signed app with its widget.                               |

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